

Doctoral Researcher,
Department of Mechanical Engineering,
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Research interests

- Abrasive based Finishing Processes
- Precision Finishing
- Tribology and Rheology
- Wear and Surface Engineering
- Environmentally Benign Coolants and Lubricants
- Feed stock development for Additive Manufacturing
- Machining of Natural and Ceramic Composites
- Electrodischarge Machining and Micro-Electrodischarge Machining

Education

- Doctor of Philosophy (Ph. D.) - Mechanical Engineering: Indian Institute of Technology IIT Tirupati, Andhra Pradesh - Jan. 2020 - Till date CGPA : 9.00 / 10
- Master of Technology (M. Tech) - Advanced Manufacturing: SASTRA Deemed University, Tamilnadu - Aug. 2017 - May. 2019 CGPA : 8.54 / 10
- Bachelor of Technology (B. Tech) - Mechanical Engineering: St. Ann's College of Engineering and Technology, Andhra Pradesh - Aug. 2012 - May. 2016 Percentage : 77.34 / 100

Research Experience

- Project Fellow at Indian Institute of Science (IISc) Bangalore Dec 2018 – Apr 2019
- Summer Intern at Indian Institute of Technology (IIT) Guwahati May 2018 – Jul 2018
- Summer Intern at National Institute of Technology (NIT) Meghalaya Jun 2016 – Jul 2016

Skills

1. Strong experimental skills, laboratory precision, and data management abilities.
2. Experience in SEM, TEM, XPS, Raman, Rheology, TGA, and related analytical methods.
3. Proven publication record in high-quality journals.
4. Interest in learning new experimental characterization techniques.
5. Excellent communication skills (written and spoken) in English.

Awards

1. **Best poster award in manufacturing** for the presentation titled “Development and performance evaluation of polymer-based form grinding tool for porous viscoelastic abrasive finishing additively manufactured Ferrous Alloy” in the conference International Conference on Additive Manufacturing and Characterization 2024, Organized by Department of Mechanical Engineering, Indian Institute of Technology Bombay, India.

2. [Best paper award in manufacturing](#) for the presentation titled “State of Art on the Chemical Polishing of Additive Manufactured Features” in the 3rd International Conference on Advances in Minerals, Metals, Materials, Manufacturing and Modelling (ICAM5)-2025, September 2025, National Institute of Technology Warangal, Warangal, Telangana, India.
3. [Best paper award in manufacturing](#) for the presentation titled “ Chemical polishing of Additive Manufactured Features: A Review ” in the 2nd International Conference on Recent Advancements in Design and Manufacturing (ICRADM)-2026, January 2026, Sardar Vallabhbhai National Institute of Technology Surat, Gujarat, India.
4. [Best paper award in manufacturing](#) for the manuscript titled “Laser Micro-Texturing of Silicon for reduced reflectivity” in the conference ICFAMMT 2022, Organized by Department of Mechanical and Aerospace Engineering, IITRAM Ahmedabad and Space Society of Mechanical Engineers, Space Application Centre, ISRO, Ahmedabad, India.
5. “[International Travel Support](#)” funded by Anusandhan National Research Framework, Government of India for attending a international conference in Chengdu, China.
6. Fellow of Indian Academy of Sciences “[Summer research fellowship program 2018](#)”, Funded by Indian Academy of Sciences, Indian National Science Academy, The National Academy of Sciences, Government of India.

Patents (Under Review)

1. M. Ravi Sankar, G. Kishor Kumar, B. Arun Kumar, [M. Shanmuka Srinivas](#), (2024), “Method of preparing phosphonium-based ionic liquid as environment-friendly cutting fluid and application thereof”. Indian Patent Filed. Publication no. IN202441075622 A1.

Journal Publications (Under Review)

1. [M. Shanmuka Srinivas](#), K. Kartheek, S. Purushothaman, N. Venkaiah, M. Ravi Sankar, “Finishing of bioresorbable Magnesium alloy by viscoelastic polymer blended flexible abrasive tool”, Journal of Materials Engineering and Performance, Second revision submitted and under review.

Journal Publications (Cumulative Impact Factor of 70.12)

1. [M. Shanmuka Srinivas](#), M. Ravi Sankar, N. Venkaiah, “Surface Integrity Improvement by Porous Viscoelastic Finishing of AM Stainless Steel”, Materials and Manufacturing Processes (Impact Factor: 7.4), 1–13. [Link](#)
2. [M. Shanmuka Srinivas](#), Abhishek Punia, T. S. Srivatsan, N. Venkaiah, M. Ravi Sankar (2026), “Tool wear in milling of metallic Materials: Types, Characteristics, and Impact”, International Journal of Metallic Materials, 1(4), 163–188. [Link](#)
3. [M. Shanmuka Srinivas](#), K. Roja, K. K. Gajrani, G. Bolar, G. Bajargan, S. M. Maliyekkal, J. Katiyar, and M. R. Sankar (2026). “Indigenous biodegradable cutting fluid effect on workpiece surface morphology and tool wear during machining of resorbable magnesium alloy”. Tribology International (Impact Factor: 6.9), 214, 111211. [Link](#)
4. D. Deepak Singh, [M. Shanmuka Srinivas](#), S. Purushothaman, M. Ravi Sankar, Nagahanumaiah, (2025) “Optimisation of Laser Parameters for Fabrication of Microchannels on PMMA using Kr-F Pulsed Laser”, Journal of Optics and Laser Technology, (Impact Factor: 5) [Link](#)
5. S. Purushothaman, [M. Shanmuka Srinivas](#), N. Venkaiah, T. S. Srivatsan, M. Ravi Sankar, (2025). “Laser surface interaction of biometals: current prospects and future trends”. International Journal of Metallic Materials, Pages 1–26. [Link](#)

6. D. Deepak Singh, N. Gouthami, Rajkumar, [M. Shanmuka Srinivas](#), S. Purushothaman, M. Ravi Sankar, Nagahanumaiah, J. Ramkumar, (2025) "Influence of Channel Geometry on the Mixing Performance in Microchannels using Non-Isothermal Liquid Streams", Journal of Micromanufacturing, (Impact Factor: 2.68), [Accepted](#).
7. [M. Shanmuka Srinivas](#), Shyam, M. Ravi Sankar, O. P. Khatri, A. Udayakumar, (2024) "Machining of Cf/SiC composite with indigenously developed nano green cutting fluid: Machined surface fibre morphology", Journal of Manufacturing Processes. (Impact Factor: 6.80) [Link](#)
8. [M. Shanmuka Srinivas](#), D. Pandey, T. Siva Prasad, S. Purushothaman, M. Ravi Sankar, (2024) "Comparative tribological studies of additive manufactured as-built and chemically polished AISi10Mg alloy surfaces". Proceedings of the Institution of Mechanical Engineers, Part J: Journal of Engineering Tribology. Volume 239(6), Pages 789-806. (Impact Factor: 1.80) [Link](#)
9. M. Satyadev, S. Purushothaman, [M. Shanmuka Srinivas](#), B. Lakshmana Rao, M. Ravi Sankar, (2024) "Numerical modelling and simulation studies of laser beam heating parameters on depth of cut and surface temperature in laser-assisted machining of Ti6Al4V", Journal of Micromanufacturing. (Impact Factor: 2.68) [Link](#)
10. [M. Shanmuka Srinivas](#), R. Meenakshi Reddy, B. Praveen Kumar, D. Bandhu, M. Das, D. Kishore, M. R. Choudhury, (2024) "Advancing microchannel fabrication: Tool-innovated die sink electrodischarge machining", Journal of Engineering. (Impact Factor: 2.30) [Link](#)
11. M. R. Choudhury, H. Dutta, [M. Shanmuka Srinivas](#), K. Debnath, T. Arunkumar, V. Upadhyay, (2024) "Optimizing drilling parameters for fly ash-filled hemp/epoxy composites: Investigating drill bit geometries, feed rates, and spindle speeds", Materials Research Express. (Impact Factor: 2.20) [Link](#)
12. [M. Shanmuka Srinivas](#), D. Kumar, P. S. Suvin, S. V. Kailas, S. R Pawar, M. R. Choudhury, (2024) "Green lubricants in action: A comprehensive performance evaluation of groundnut oil-based cutting fluids in metal machining processes", Physica Scripta. (Impact Factor: 2.60) [Link](#)
13. P. Sharma, S. Purushothaman, [M. Shanmuka Srinivas](#), M. Sachdeva, N. Venkaiah, M. Ravi Sankar (2024) "Experimental study and modelling of laser micro texturing of Nitinol", Journal of Micromanufacturing. (Impact Factor: 2.68) [Link](#)
14. T. Siva Prasad, [M. Shanmuka Srinivas](#), M. Ravi Sankar, (2023) "Sustainability aspects of composite filament fabrication for 3D printing applications", Renewable and Sustainable Energy Reviews, Volume 189, Pages 113961. (Impact Factor: 16.30) [Link](#)
15. S. Purushothaman, [M. Shanmuka Srinivas](#), N. Venkaiah, M. Ravi Sankar, (2023) "Salient aspects on micro-electrical discharge machining of nitinol for cardiovascular applications", Journal of Micromanufacturing. (Impact factor: 2.68) [Link](#)
16. M. Satyadev, [M. Shanmuka Srinivas](#), J. K. Katiyar, B. Lakshmana Rao, and M. Ravi Sankar, (2023) "State-of-the-art hybrid-assisted machining processes of nickel-based superalloys", Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering. (Impact Factor: 2.20) [Link](#)
17. M. Sreekumar, S. Purushothaman, [M. Shanmuka Srinivas](#), J. K. Katiyar, and M. R. Sankar, (2022) "A review of additives in abrasive water jet machining and their performance", Proceedings of the Institution of Mechanical Engineers, Part J: Journal of Engineering Tribology. (Impact Factor: 1.80) [Link](#)
18. M. R. Choudhury, [M. Shanmuka Srinivas](#), K. Debnath, (2018) "Experimental investigations on drilling of lignocellulosic fiber reinforced composite laminates", Journal of Manufacturing Processes, Volume 34, Pages 51-61. (Impact Factor: 6.80) [Link](#)

Procedia

1. [M. Shanmuka Srinivas](#), M. R. Sankar, N. Venkaiah, (2025) "Experimental Investigation on Post Processing of Additively Manufactured Surgical Steel by Viscoelastic Abrasive Finishing. [Link](#)
2. [M. Shanmuka Srinivas](#), S. Purushothaman, N. Venkaiah, M. R. Sankar, (2023) "State of the art on tool wear characterization in micro-milling", Materials Today: Proceedings. (Impact factor: 2.5) [Link](#)
3. [M. Shanmuka Srinivas](#), K. K. Gajrani, A. Udayakumar, M. R. Sankar, (2021) "Sustainable Machining of Cf/SiC Ceramic Matrix Composite Using Green Cutting fluids", Procedia CIRP, Volume 98, Pages 151-156. (Impact Factor: 2.1) [Link](#)

Conference Presentations (Oral)

1. [M. Shanmuka Srinivas](#), M. Ravi Sankar, N. Venkaiah, "Surface Improvement by Porous Viscoelastic Finishing of Additive Manufactured Stainless Steel", 7th International Conference on Applied Surface Science (ICASS-2026), 1-4 June 2026, Chengdu, China.
2. S. Purushothaman, [M. Shanmuka Srinivas](#), M. Ravi Sankar, N. Venkaiah, "Experimental investigation of surface enhancement in nitinol stents using pulsed laser micro polishing", 7th International Conference on Applied Surface Science (ICASS-2026), 1-4 June 2026, Chengdu, China.
3. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, Venkata Subbaiah G, M. Ravi Sankar, "Chemical polishing of Additive Manufactured SS 316L Lattice Structures", 2nd International Conference on Additive Manufacturing and Characterization 2026 (ICAMC'26), February 2026, Indian Institute of Technology Madras, Chennai India.
4. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "Chemical polishing of Additive Manufactured Features: A Review ", 2nd International Conference on Recent Advancements in Design and Manufacturing (ICRADM)-2026, January 2026, Sardar Vallabhbhai National Institute of Technology Surat, Gujarat, India.
5. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "The effect of dynamic chemical polishing on additive manufactured features", 2nd International Conference on Advanced Functional Materials and Devices (AFMD-2026), March 2026, Nanotechnology Research Centre (NRC), SRM Institute of Science and Technology, Kattankulathur, Tamilnadu, India.
6. [M. Shanmuka Srinivas](#), N. Venkaiah, M. Ravi Sankar, "Experimental investigation on comparative tribological aspects of as-built and viscoelastic polymer-based abrasive finished surgical steel surfaces", 13th International Conference on Tribology, IndiaTrib 2025 December 2025, Indian Institute of Technology Bhilai, Chattisgarh, India.
7. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "Surface characteristics of Additive Manufactured SS316L Lattice Structures", 13th International Conference on Tribology, IndiaTrib 2025 December 2025, Indian Institute of Technology Bhilai, Chattisgarh, India.
8. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "Dynamic Chemical polishing of Additive Manufactured SS316L Primitive Lattice Structures", 10th International and 31st All India Manufacturing Technology, Design and Research (AIMTDR 2025) Conference, December 2025, Indian Institute of Technology Indore, Indore, Madhya Pradesh, India.
9. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "Chemical Polishing of Additive Manufactured Features", 3rd International Conference on Advances in Minerals, Metals, Materials, Manufacturing and Modelling (ICAM5)-2025, December 2025, SRM UNIVERSITY, Amaravati, Andhra Pradesh, India.

10. [M. Shanmuka Srinivas](#), N. Venkaiah, M. Ravi Sankar, "Enhancing the surface integrity of additively manufactured Nickel-based alloy using novel polymer based viscoelastic abrasive finishing process" APA International conference on Polymers for Advanced Technology (APA 2025), October 2025, Udaipur, India.
11. [M. Shanmuka Srinivas](#), N. Venkaiah, M. Ravi Sankar, "Experimental Investigation on Post Processing of Additively Manufactured Surgical Steel by Viscoelastic Abrasive Finishing (ASME IMECE 2025), September 2025, Hyderabad, India.
12. [M. Shanmuka Srinivas](#), P. Vinod Kumar, M. Gururaj Acharya, M. Ravi Sankar, "Surface Finishing of Biodegradable Magnesium Alloy (ZE41A) using Viscoelastic Polymer Abrasive Tool : Effect of Interfacial Friction and Abrasion", International Conference on Friction based Processes (ICFP 2025), September 2025, Indian Institute of Technology Tirupati, Tirupati, India.
13. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "State of Art on the Chemical Polishing of Additive Manufactured Features", 3rd International Conference on Advances in Minerals, Metals, Materials, Manufacturing and Modelling (ICAM5)-2025, September 2025, National Institute of Technology Warangal, Warangal, Telangana, India.
14. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "State of Art on the Tribological Aspects of Additive Manufactured Features", 3rd International Conference on Advances in Minerals, Metals, Materials, Manufacturing and Modelling (ICAM5)-2025, September 2025, National Institute of Technology Warangal, Warangal, Telangana, India.
15. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "State of art on the characteristics of additive manufactured lattice structures", International Conference On Advanced Materials and Characterization (ICAMC 2025), August 2025, Sathyabama Institute of Science and Technology, Chennai, Tamil Nadu, India.
16. S. Purushothaman, [M. Shanmuka Srinivas](#), K. K. Gajrani, N. Venkaiah, M. Ravi Sankar, "Surface Refinement of Selective Laser Sintered AlSi10Mg using Laser Polishing", International Conference and Workshop on Additive Manufacturing and Characterization (ICAMC 2025), January 2025, Indian Institute of Technology Bombay, India.
17. [M. Shanmuka Srinivas](#), D. Pandey, T. Siva Prasad, P. Sangeeth, M. Ravi Sankar, "Experimental investigation on tribological studies of as-built and chemical polished surgical steel surfaces", International Conference on Tribocorrosion (ICTC 2024), December 2024, Indian Institute of Technology Delhi, Delhi, India.
18. [M. Shanmuka Srinivas](#), S. P. Mohesvari, G. Kethana Priya, M. Ravi Sankar, "Experimental Investigation on the Rheological Behaviour of Hydroxylated Hexagonal Boron Nitride Nanoparticles Dispersed in Castor Oil", International Conference on Precision, Meso, Micro and Nano Engineering (COPEN 2024), December 2024, National Institute of Technology Calicut, Calicut, India.
19. [M. Shanmuka Srinivas](#), P. Vinod Kumar, G. Acharya, P. Sangeeth, N. Venkaiah, M. Ravi Sankar, "Precision Finishing of ZE41A Magnesium Alloy with Viscoelastic Polymer Blended Abrasives", International Conference on Precision, Meso, Micro and Nano Engineering (COPEN 2024), December 2024, National Institute of Technology Calicut, Calicut, India.
20. [M. Shanmuka Srinivas](#), S. Purushothaman, N. Venkaiah, M. Ravi Sankar, "Development and performance evaluation of polymer-based abrasive tool for porous viscoelastic abrasive finishing additively manufactured SS 316L", International Conference on Advances in Foundry and Forming Technology (ICAFFT 2024), December 2024, National Institute of Advanced Manufacturing Technology, Ranchi, India.

21. M. Shanmuka Srinivas, P. Vinod Kumar, N. Venkaiah, M. Ravi Sankar, "Viscoelastic Polymer Blended Abrasive Finishing of Biodegradable Magnesium alloy", International Conference on Biotechnology, Medicine and Cancer Therapeutics-2024 (ICBMCT 2024), December 2024, Sri Venkateswara University, Tirupati, India.
22. M. Shanmuka Srinivas, S. Purushothaman, N. Venkaiah, M. Ravi Sankar, "Experimental Investigation on Surface Finishing of Additively Manufactured SS 316L using Visco-elastic Porous Abrasive Tool", International Conference on next generation technologies (ICNGT 2024), November 2024, Indian Institute of Technology Madras, Chennai, India.
23. M. Shanmuka Srinivas, K. Kartheek, S. Purushothaman, N. Venkaiah, M. Ravi Sankar, "Finishing of Bioresorbable Magnesium Alloy by Viscoelastic Polymer Blended Minimal Geometrically Deformable Abrasive Tool", 9th International and 30th All India Manufacturing Technology, Design and Research (AIMTDR - 30) Conference, December 2023, Indian Institute of Technology Banaras Hindu University, Varanasi, India.
24. M. Shanmuka Srinivas, S. Purushothaman, N. Venkaiah, M. Ravi Sankar, "Experimental Investigation on the Burnishing of Additively Manufactured Thin-Walled Stainless Steel", ASME AM 3D AERO 2023, December 2023, Ramaiah institute of Technology, Bangalore, India.
25. S. Purushothaman, M. Shanmuka Srinivas, M. Mahaboob Basha, K. Bobe Bonsa, M. Ravi Sankar, "Experimental Study of Chemical Assisted Laser Beam Surface Finishing of Silicon Carbide", ASME AM 3D AERO 2023, December 2023, Ramaiah institute of Technology, Bangalore, India.
26. S. Purushothaman, M. Shanmuka Srinivas, M. Satyadev, N. Venkaiah, M. Ravi Sankar, "Salient aspects on Laser Assisted Bioprinting", International Conference On Material Processing Using Lasers and Surface Engineering (IMPULSE), December 2023, Indian Institute of Technology Madras, Chennai, India.
27. M. Shanmuka Srinivas, S. Purushothaman, N. Venkaiah, M. Ravi Sankar, "State of the art on tool wear characterization in micro-milling", Indo-European Conference on Advanced Manufacturing and Materials Processing (IECAdvaMAP-2023), February 2023, Alappuzha, Kerala, India.
28. S. Purushothaman, M. Shanmuka Srinivas, M. Ravi Sankar, "Laser micro-texturing of silicon for reduced reflectivity", International Conference on Futuristic Advancements in Materials, Manufacturing and Thermal Sciences (ICFAMMT 2022), January 2022, Infrastructure Technology Research and Management, Ahmedabad, India.
29. Shyam, M. Shanmuka Srinivas, K. Kumar Gajrani, A. Udayakumar, M. Ravi Sankar, "Sustainable Machining of Cf/SiC Ceramic Matrix Composite Using Green Cutting Fluids", The 28th CIRP Conference on Life Cycle Engineering, March, 2021, Jaipur, India.
30. Jaspreet Singh, M. Shanmuka Srinivas, V. Surya Ganesh, P. Venkata Kedara Krishna, M. Ravi Sankar, "Experimental investigation and modelling on the end milling of wire arc additive manufactured feature", 2nd Virtual International Tribology Research Symposium (ITRS2021), December, 2021.
31. B. Sai Teja, M. Shanmuka Srinivas, M. Mahaboob Basha, M. Ravi Sankar, "Tribological Study of Additively Manufactured Polymers for Simple to Complex Features: A critical review", 2nd Virtual International Tribology Research Symposium (ITRS2021), December, 2021.

Conference Presentations (Poster)

1. M. Shanmuka Srinivas, N. Venkaiah, M. Ravi Sankar, "Process-Surface Correlation in Viscoelastic Polymer Abrasive Form Finishing of Additively Manufactured Surgical Stainless Steel", Research scholars day - 2026, Indian Institute of Technology Tirupati, January 2026.

2. Deepak Singh, [M. Shanmuka Srinivas](#), P. Sangeeth, M. Ravi Sankar, Nagahanumaiah, "Investigation of flow dynamics and mixing efficiency in multi-cell micro tesla valves", APA International conference on Polymers for Advanced Technology (APA 2025), October 2025, Udaipur, India.
3. [M. Shanmuka Srinivas](#), S. Purushothaman, N. Venkaiah, M. Ravi Sankar, "Development and performance evaluation of polymer-based form grinding tool for porous viscoelastic abrasive finishing additively manufactured Ferrous Alloy", International Conference and Workshop on Additive Manufacturing and Characterization (ICAMC 2025), January 2025, Indian Institute of Technology Bombay, India.
4. A. Dhananjayulu, [M. Shanmuka Srinivas](#), M. Satyanarayana, M. Ravi Sankar, "State of art on the Surface characteristics of Additive Manufactured Lattice Structures", International Conference on Advanced Materials and Characterization (ICAMC 2025), August 2025, Sathyabhama Institute of Technology, India.

Book Chapters

1. B. D. K. Patro, [M. Shanmuka Srinivas](#), P. S. Suvin and M. R. Sankar, (2025), "Nanocutting Fluids for Machining", Manufacturing Tribology, CRC Press. [Link](#)
2. T. Siva Prasad, [M. Shanmuka Srinivas](#), D. Venkata Krishna, P. Bhargav Chandan, M. Ravi Sankar, (2025), "Various Challenges in Metal and Ceramic Particle-Filled Composite Filament-Based 3D Printing: A Review", Manufacturing Strategies and Systems, CRC Press. [Link](#)
3. S. Purushothaman, [M. Shanmuka Srinivas](#), R. R. Behera, N. Venkaiah, M. R. Sankar, (2021), "Laser micro-texturing of silicon for reduced reflectivity", Sustainable Materials and Manufacturing Technologies, CRC Press. [Link](#)
4. [M. Shanmuka Srinivas](#), R. Panneer, P. S. Suvin and S. V. Kailas, (2021), "Synthesis and Testing of Novel Neem Oil-Based Cutting Fluid with Ionic Liquid Additives", Industry 4.0 and Advanced Manufacturing, Springer. [Link](#)

Positions of Responsibility

1. Teaching assistance for the courses at IIT Tirupati: [Abrasive Machining and Finishing Processes](#), [Machining Processes](#), [Engineering Drawing](#), [Product Design and Development](#), [Metrology Laboratory](#)
2. Teaching assistance for [National Program on Technology Enhanced Learning \(NPTEL\)](#) online course: [Introduction to Machining and Machining Fluids](#).